

POKHARA UNIVERSITY

Level: Bachelor

Semester: Fall

Year: 2024

Programme: BCSIT

Full Marks: 100

Course: Fundamentals of Computer System

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. [10×2]

1. List any four characteristics of a computer.
2. Define primary memory. Why was cache memory introduced?
3. Differentiate between system development software and system management software.
4. Write about open source operating system.
5. Define Computer Network. What are the criteria of good network?
6. What does IP stand for? Differentiate between IPv4 and IPv6.
7. Define DBMS.
8. Define Virtual Reality.
9. What is cryptography?
10. What is IOT? List the components of an IoT system.

Section "B"

Descriptive Answer Questions

Attempt any five questions. [5×10]

11. What is the CPU Scheduling function of an operating system? Explain any two algorithms used for CPU Scheduling.
12. Explain simplex, half-duplex, and full-duplex communication modes with examples of their applications.
13. What is a packet in Internet communication? Describe the working mechanism of the Internet, including the roles of ISPs, routers, and packets in data transmission.
14. What are the elements of multimedia? Explain the importance of multimedia in web-based applications.
15. What are ethical issues related to computer security? Briefly describe spyware, malware, and ransomware.
16. What is Business Intelligence (BI)? Discuss the tools and techniques used in BI, such as data mining, data visualization, and dashboards, and how they help businesses improve performance.

Section "C"
Long Answer Questions

Attempt **any two** questions. [2×15]

17. Explain the classification of computer on the basis of purpose, data, size, brands and models.
18. What is DBMS? Compare the database approach with the file system approach. Discuss the characteristics and functions of RDBMS with examples.
19. Explain why the security mechanism of block chain technology is robust, using crypto currency as an example.